

PortPulse

Container Congestion
Predictor & Port Operations
Optimiser

TEAM MEMBERS

- ♦ Jigar Sakhiya
- ♦ Trusha Patel
- ♦ Hetvi Tank
- ♦ Priyansh Sukhadiya



THE CHALLENGE

The Port Congestion Problem

In 2021, the ports of Los Angeles and Long Beach experienced severe congestion, triggering widespread delays and supply-chain disruption.

— The Core Operational Challenges

- Manual, spreadsheet-based planning that cannot keep pace
- Reactive congestion detection instead of early warning
- Vessels left waiting offshore with no clear berth window
- Difficulty making timely alternate-port decisions
- Lack of a unified, real-time operational view



Who Is Affected & Why It Matters

KEY USERS

- Port operation managers
- Shift supervisors
- Vessel planners
- Terminal operators
- Shipping & logistics coordinators

CORE PROBLEMS

- Manual, spreadsheet-based planning
- Reactive congestion detection
- Vessels left waiting offshore
- Hard alternate-port decisions
- No unified operational view

BUSINESS IMPACT

- Longer waiting & lower utilisation
- Higher costs and delivery delays
- Increased emissions from idling
- Reduced end-to-end visibility

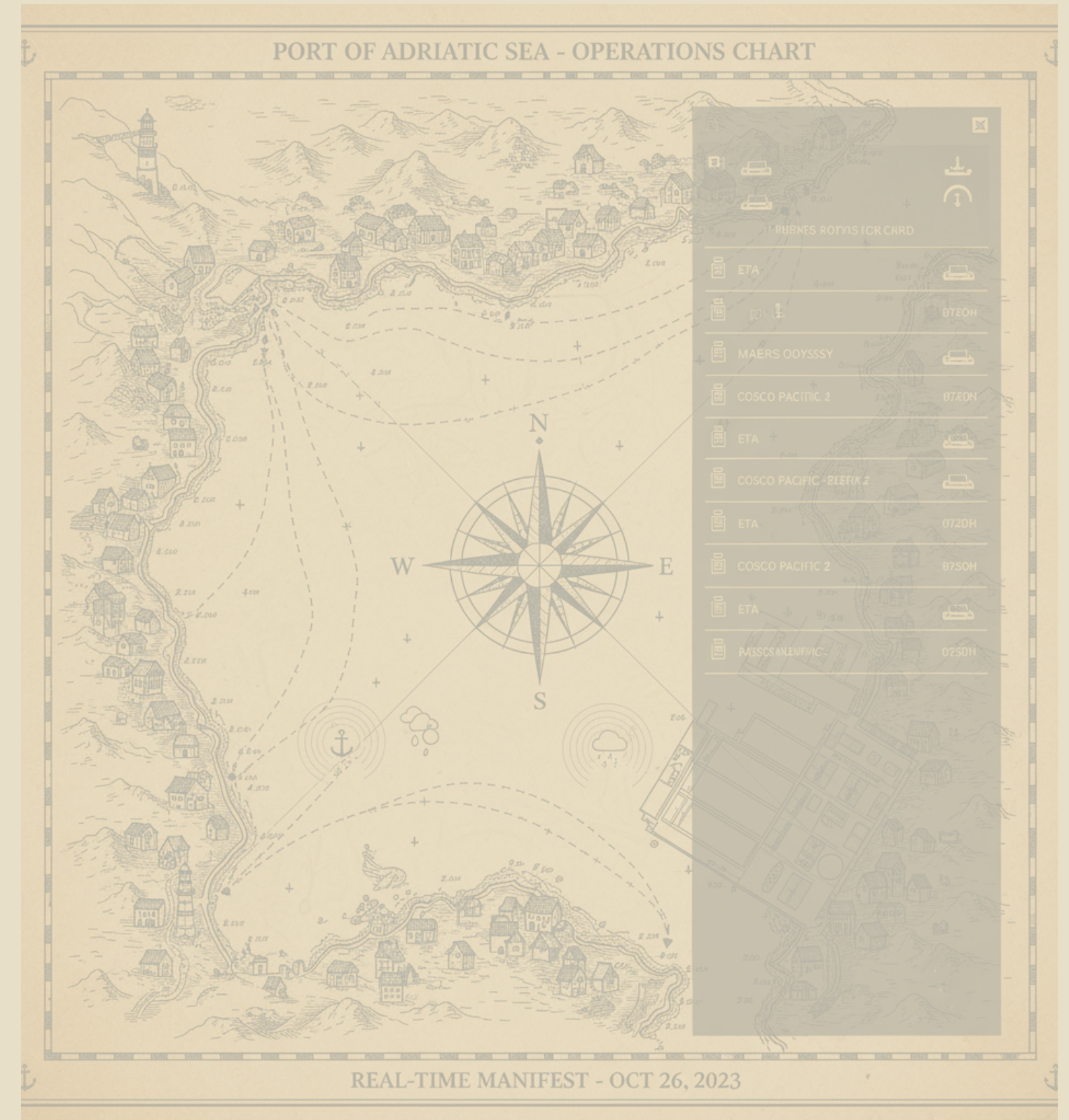


OUR SOLUTION

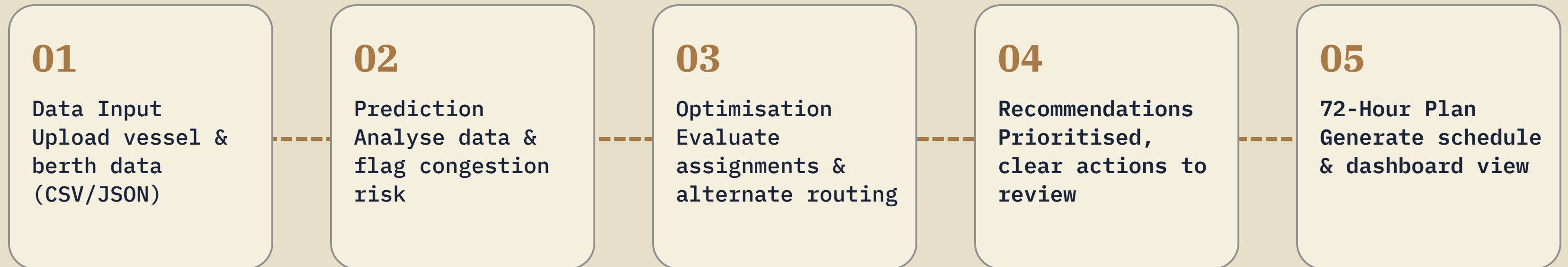
An AI-Assisted Port Operations Planning Platform

PortPulse turns fragmented port data into clear, prioritised recommendations – using deterministic rule-based planning enhanced by AI reasoning.

- Predicts congestion hotspots before they escalate
- Identifies overloaded berths and conflicts
- Recommends alternate routing options
- Optimises vessel-to-berth assignments
- Produces a clear 72-hour operational plan



How PortPulse Works



A stepwise pipeline: upload → analysis → congestion risk prediction → assignment evaluation → alternate routing comparison → plan generation → dashboard display.

Product / Demo Walkthrough



01

Vessel & Berth Data Input

UPLOAD VESSEL & BERTH DATASETS VIA CSV /
JSON WITH STRICT INPUT VALIDATION.

02

Congestion Risk Dashboards

COLOR-CODED CONGESTION RISK DASHBOARDS FLAG
HOTSPOTS BEFORE THEY ESCALATE.

03

Recommended Assignments

PRIORITISED VESSEL-TO-BERTH ASSIGNMENTS
RESOLVE CONFLICTS AUTOMATICALLY.

04

Alternate-Port Suggestions

ALTERNATE-PORT SUGGESTIONS WITH ROUTE MAPS
FOR REROUTING DECISIONS.

05

72-Hour Operational Schedule

A CLEAR 72-HOUR OPERATIONAL SCHEDULE READY
FOR SHIFT HANDOVER.

06

Operational Summary (IBM AI)

OPERATIONAL SUMMARY GENERATED BY IBM AI FOR
SUPERVISOR REVIEW.

Technical Architecture

API LAYER

FastAPI service exposing endpoints,
with clean separation of API and
domain logic.

DOMAIN MODULES (Python)

prediction.py · routing.py ·
assignment.py · planner.py ·
summary.py · chat.py

DATA & DELIVERY

CSV/JSON ingestion · static web UI
· automated tests · Docker support

DESIGN PRINCIPLES

- Strict input validation on all data
- Rule-based planning across vessel
schedules, berth capacity &
conflicts
- Deterministic logic built for fast
decision support
- Clear API / domain separation for
maintainability
- Sample CSV data today, with a
defined path to a live system

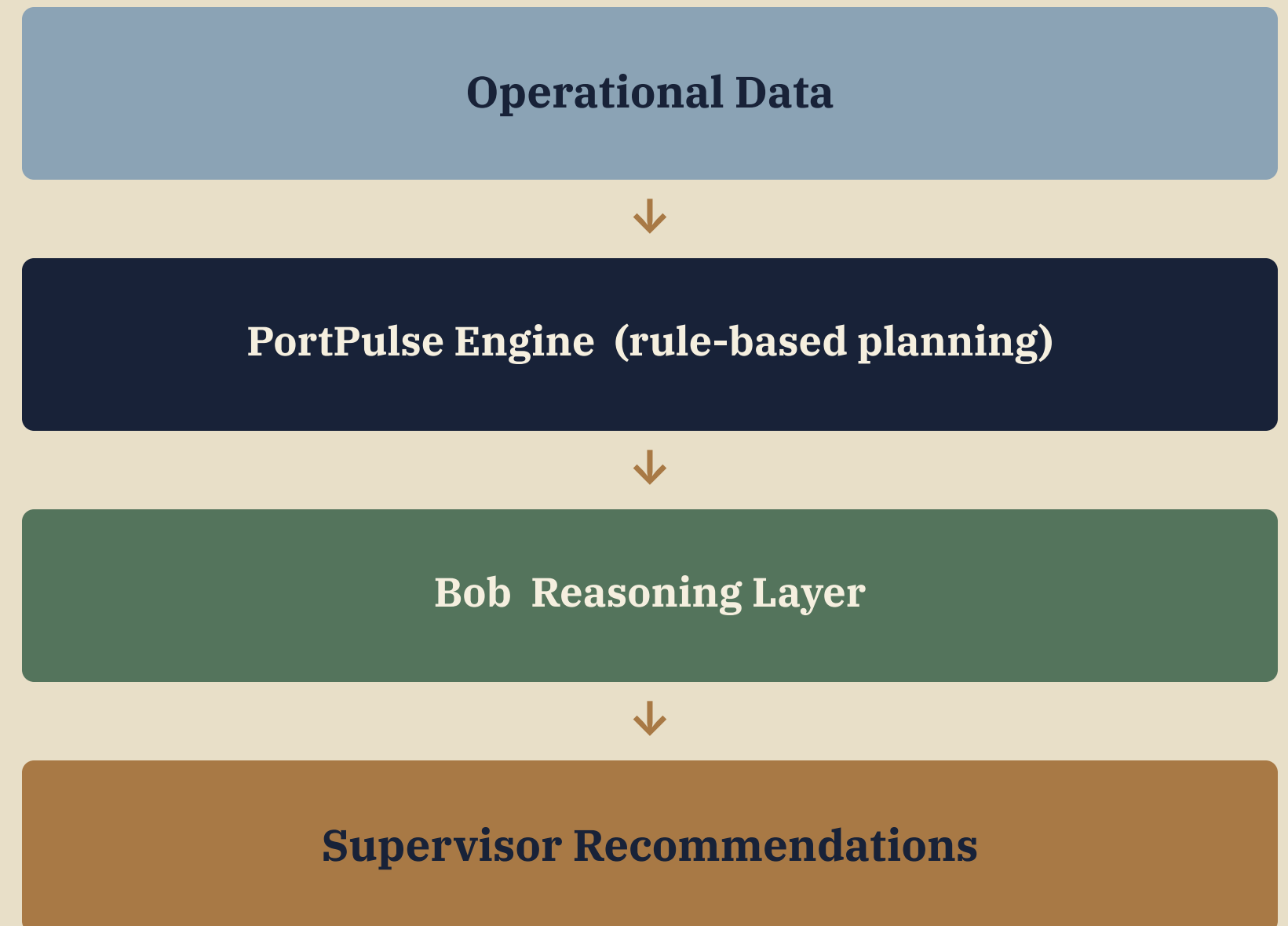
PROTOTYPE SCOPE

This is a hackathon
prototype using AI-
assisted, rule-based
optimisation – not
advanced ML or full
production-scale, real-
time systems. The
architecture is built to
upgrade toward live
operation.

Bob Integration

AI reasoning layered on top of deterministic planning – with humans always in control.

- Natural-language explanation of congestion predictions
- Summarisation of plans and operational status
- Q&A on priorities and next actions
- Alternate-port strategy comparisons
- Clear operational recommendations



Impact & Future Potential

IMMEDIATE BENEFITS

- Earlier congestion visibility
- Faster, better-informed decisions
- Improved berth utilisation
- Fewer delays across operations
- Consistent shift handovers
- Clearer team communication

FUTURE OPPORTUNITIES

- Real-time AIS + weather / tide / labor data
- Crane and yard optimisation
- Longer forecasting horizons
- Historical data learning
- What-if scenario simulation
- Multi-port coordination
- KPI tracking: waiting time, utilisation, emissions, avoided congestion

PORTPULSE

From reactive queue management to proactive, AI-assisted decision-making.

PortPulse gives port teams the foresight to act early – turning congestion data into clear, confident decisions.

Team Surfers